# Design Considerations for the RC-Coupled BJT Amplifier

Q-Point Placement, Gain Equations, and the Early Effect

Companion to *Frequency Response of the RC-Coupled BJT Amplifier*.
Same circuit, same bias philosophy — now we move the operating point and watch what breaks.

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**What this document does**

We start from the textbook design rule $V_{CE} = V_{CC}/2$, build the baseline amplifier, and then deliberately move the Q-point around. At every position we ask four questions: *What is the gain? How much can the output swing before clipping? How much power does it burn? And what does the frequency response do?*

Every gain expression is given twice — once with the transistor treated as an ideal current source, and once including the **Early effect** (the finite output resistance $r_o$). We will find a simple rule for exactly how much error the ideal treatment costs, and identify the one situation where ignoring $r_o$ is catastrophic rather than harmless.

## Contents

# 1 Ground Rules: What Is Held Fixed

To make comparisons meaningful, the **DC biasing method** is identical for every design in this document. Only two numbers are ever allowed to change: the collector current $I_C$ and the collector resistor $R_C$.

| **Held fixed** | **Value** | **Why** |
|---|---|---|
| Supply | $V_{CC} = 12\,\text{V}$ | given |
| Base voltage | $V_B = 0.2\,V_{CC} = 2.4\,\text{V}$ | divider ratio $R_1{:}R_2 = 4{:}1$ |
| Emitter voltage | $V_E = V_B - 0.7 = 1.7\,\text{V}$ | $\geq 1\,\text{V}$ for bias stability |
| Divider stiffness | $I_{div} = 21\,I_B$ | so $V_B$ is independent of $\beta$ |
| Load, source | $R_L = 4\,\text{k}\Omega$, $R_s = 600\,\Omega$ | given |
| Device | $\beta = 100$, $V_A = 100\,\text{V}$ | $V_T = 26\,\text{mV}$ |
| Capacitors | $C_{C1} = C_{C2} = 1\,\mu\text{F}$, $C_E = 20\,\mu\text{F}$ | from the companion document |
| | $C_\mu = 4\,\text{pF}$, $C_{je} = 8\,\text{pF}$, $\tau_F = 0.5\,\text{ns}$ | so $C_\pi = C_{je} + g_m\tau_F$ |
| **Free** | $I_C$ and $R_C$ | the two design knobs |

Because $V_E$ is fixed, the emitter resistor follows immediately from the chosen current, and the divider scales to stay equally stiff:

$$R_E = \frac{V_E}{I_C} = \frac{1.7\,\text{V}}{I_C}, \qquad R_1 + R_2 = \frac{V_{CC}}{21\,I_B} = \frac{12\beta}{21\,I_C}, \qquad R_1 = 0.8(R_1{+}R_2). \tag{1}$$

**The one equation that governs the whole design space**

Walk around the collector–emitter loop, using $I_E \approx I_C$:

$$V_{CC} = \underbrace{I_C R_C}_{V_{RC}} + V_{CE} + \underbrace{I_C R_E}_{V_E = 1.7\,\text{V}} \quad \Longrightarrow \quad \boxed{V_{RC} + V_{CE} = V_{CC} - V_E = 10.3\,\text{V}} \tag{2}$$

The DC voltage dropped across $R_C$ and the DC voltage held across the transistor **add up to a constant**. Every design decision in this document is a decision about how to split that 10.3 V. As we are about to see, $V_{RC}$ buys *gain* and $V_{CE}$ buys *swing headroom*, so Eq. (2) is a conservation law for the central trade-off of the circuit.

# 2 The Baseline Design: $V_{CE} = V_{CC}/2$

## 2.1 Design procedure

The classical rule places the Q-point at mid-supply so that the collector can swing equally far in both directions. Applying it with $V_E$ already fixed at 1.7 V:

1. Choose $V_{CE} = V_{CC}/2 = 6\,\text{V}$. From Eq. (2), $V_{RC} = 4.3\,\text{V}$.
2. Choose a bias current. Taking $I_C = 1.13\,\text{mA}$ (a compromise between power and gain), $R_E = 1.7/1.13\text{m} = 1.5\,\text{k}\Omega$.
3. $R_C = V_{RC}/I_C = 4.3/1.13\text{m} = 3.79\,\text{k}\Omega$. The nearest standard value is $3.9\,\text{k}\Omega$; we use $\mathbf{4\,k\Omega}$ to match the companion document exactly, which gives $V_{CE} = 5.77\,\text{V}$ — within 4% of mid-supply.
4. Divider: $I_B = 11.3\,\mu\text{A}$, so $R_1 + R_2 = 12/(21 \times 11.3\mu) = 50\,\text{k}\Omega$, split 4:1 as $R_1 = 40\,\text{k}\Omega$, $R_2 = 10\,\text{k}\Omega$.

This is exactly the amplifier analysed in the companion document. We call it **Design A**.

## 2.2 The two load lines

The Q-point sits on the **DC load line**, whose slope is set by $R_{dc} = R_C + R_E$. But the *signal* does not see $R_E$ (it is bypassed) and does see $R_L$ (coupled in through $C_{C2}$). So the signal excursion follows a steeper **AC load line** through the same Q-point, with slope set by

$$R_{ac} = R_C \parallel R_L = 4 \parallel 4 = 2\,\mathrm{k\Omega}, \qquad R_{dc} = R_C + R_E = 5.5\,\mathrm{k\Omega}. \tag{3}$$

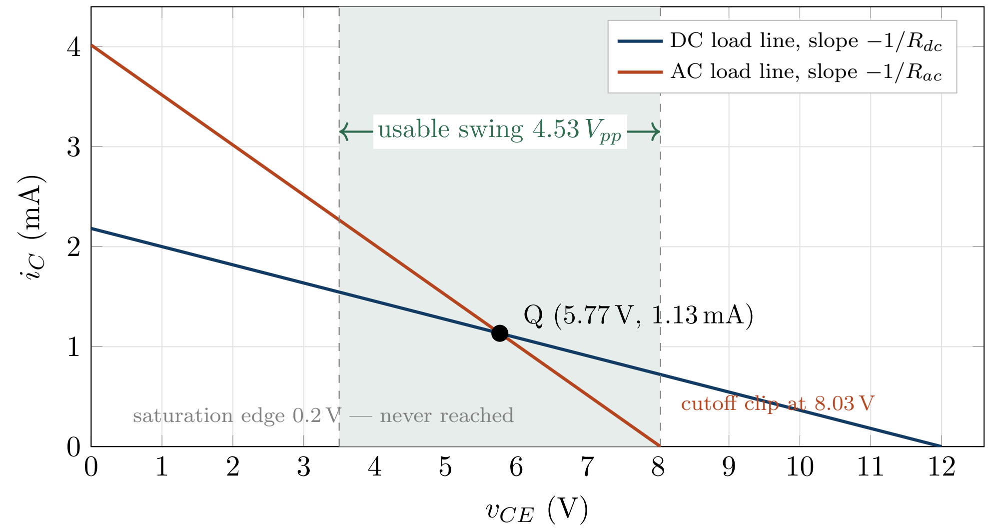


Figure 1: DC and AC load lines for Design A. The signal rides the *AC* line, which is much steeper. The output can only rise 2.27 V before the transistor cuts off, but could fall 5.57 V before it saturates — so the swing is badly asymmetric and cutoff-limited.

The two clipping limits are

$$\underbrace{v_o^+ = I_C R_{ac}}_{\text{cutoff limit}} = 1.133\text{m} \times 2\text{k} = 2.27\,\mathrm{V}, \qquad \underbrace{v_o^- = V_{CE} - V_{CE,sat}}_{\text{saturation limit}} = 5.77 - 0.2 = 5.57\,\mathrm{V}, \tag{4}$$

and the undistorted output is limited by the smaller of the two: $V_{pp} = 2\min(v_o^+, v_o^-) = 4.53\,\mathrm{V}$.

**Mid-supply is the wrong rule when an AC-coupled load is present**

$V_{CE} = V_{CC}/2$ centres the Q-point on the *DC* load line. But the signal travels on the *AC* load line, and because $R_{ac} < R_{dc}$, that line is steeper — so a Q-point centred on the DC line is *not* centred on the AC line. Design A wastes 3.3 V of headroom on the saturation side that it can never use, while clipping early on the cutoff side.
The Q-point that centres the *AC* load line satisfies $I_C R_{ac} = V_{CE} - V_{CE,sat}$. Substituting $I_C = (V_{CC} - V_{CE})/R_{dc}$ and solving gives

$$V_{CE,opt} \approx \frac{V_{CC}\, R_{ac}}{R_{ac} + R_{dc}} \tag{5}$$

which for Design A's resistors is $12 \times 2/7.5 = 3.2\,\mathrm{V}$ — barely half of mid-supply. We build that amplifier as **Design D** in Section 5.

# 3 Gain Equations, Case by Case

### 3.1 Case 1 — ideal transistor, fully bypassed emitter

Replacing the transistor with $r_\pi$ and an *ideal* current source $g_m v_{be}$ (infinite output resistance), the collector node sees only $R_C \parallel R_L$:

$$\boxed{A_v = \frac{v_o}{v_b} = -g_m R_{ac} = -\frac{R_{ac}}{r_e}} \qquad g_m = \frac{I_C}{V_T}, \quad r_e = \frac{V_T}{I_C}, \quad r_\pi = \beta r_e. \tag{6}$$

Now substitute $g_m = I_C/V_T$ and, for the moment, drop $R_L$ so that $R_{ac} \to R_C$:

$$|A_v|_{\text{unloaded}} = \frac{I_C R_C}{V_T} = \boxed{\frac{V_{RC}}{V_T}} \tag{7}$$

**The most useful result in this document**

The unloaded voltage gain of a common-emitter stage equals **the DC voltage dropped across the collector resistor, divided by 26 mV**. It does not depend on $I_C$ or $R_C$ separately — only on their product.
For Design A: $|A_v| = 4.53/0.026 = 174$. Halved by the equal load $R_L$, this gives 87 — the number computed in the companion document.
The consequence is severe. Because Eq. (2) caps $V_{RC}$ at 10.3 V, a resistively loaded stage on a 12 V supply can never have an unloaded gain above about $10.1/0.026 \approx 390$, no matter what transistor, what current, or what resistor you choose.

With the load restored, Eq. (7) becomes

$$|A_v| = \frac{V_{RC}}{V_T} \cdot \frac{R_L}{R_C + R_L}, \tag{8}$$

which is worth staring at: for a *fixed Q-point* (fixed $V_{RC}$), you get more gain by making $R_C$ *smaller* and $I_C$ correspondingly larger, because the resistive divider between $R_C$ and $R_L$ becomes less lossy. We exploit this in Section 6.

### 3.2 Case 2 — including the Early effect

Real transistors do not have flat output characteristics. Increasing $v_{CE}$ narrows the effective base width, which increases the collector current slightly. Extrapolated backwards, the output characteristics all meet at $-V_A$, the **Early voltage**. The slope this produces is modelled as a resistor $r_o$ from collector to emitter:

$$\boxed{r_o = \frac{V_A}{I_C}} \qquad \text{Design A:} \quad r_o = \frac{100}{1.133\text{m}} = 88.2\,\text{k}\Omega. \tag{9}$$

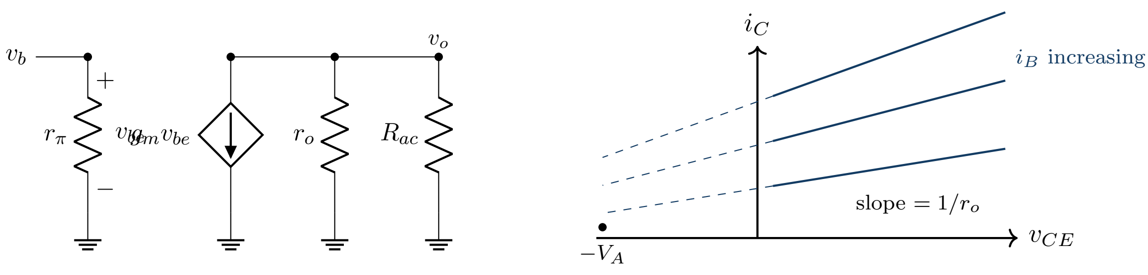


Figure 2: Left: hybrid-$\pi$ model with $r_o$ added — it simply appears in parallel with the load. Right: the Early effect is the finite slope of the output characteristics, which extrapolate to $-V_A$.

Since $r_o$ sits directly across the output node, it merely joins the parallel combination:

$$\boxed{A_v = -g_m\,(r_o \parallel R_C \parallel R_L)} \tag{10}$$

Design A: $A_v = -43.6\text{m} \times (88.2\text{k} \parallel 2\text{k}) = -43.6\text{m} \times 1.956\text{k} = \mathbf{-85.2}$ versus $-87.2$ without $r_o$.

### 3.3 How much does ignoring the Early effect cost?

Because $r_o \gg R_{ac}$, the parallel combination is barely disturbed, and a first-order expansion of $R_{ac} \parallel r_o$ gives a strikingly simple error rule:

$$\frac{|A_v|_{\text{ideal}} - |A_v|_{\text{with } r_o}}{|A_v|_{\text{with } r_o}} = \frac{R_{ac}}{r_o} = \frac{I_C R_{ac}}{V_A} = \boxed{\frac{v_o^+}{V_A}} \tag{11}$$

**A memorable way to judge whether $r_o$ matters**

$I_C R_{ac}$ is exactly the cutoff-side output swing $v_o^+$ from Section 2.2. So:

*The percentage error from ignoring the Early effect equals the available output swing expressed as a percentage of the Early voltage.*

Design A can swing 2.27 V and has $V_A = 100\,\text{V}$, so the error is 2.27% — confirmed exactly by the numbers above. Since a 12 V amplifier can never swing more than about 10 V, and modern small-signal transistors have $V_A$ between 50 and 150 V, **the Early effect in a resistively loaded discrete stage is always a correction of a few percent** — smaller than the tolerance of the resistors around it.

### 3.4 Case 3 — source loading

Neither expression above is what you measure at the source. The divider formed by $R_s$ and the amplifier input resistance $R_i$ comes first:

$$R_i = R_1 \parallel R_2 \parallel r_\pi, \qquad \boxed{A_{vs} = \frac{v_o}{v_s} = A_v \cdot \frac{R_i}{R_i + R_s}} \tag{12}$$

This term is where many otherwise sensible designs quietly lose their gain, because $r_\pi = \beta V_T / I_C$ *shrinks* as you raise the bias current. Chasing $g_m$ by raising $I_C$ therefore lowers $R_i$ at the same time.

### 3.5 Case 4 — partially bypassed (swamped) emitter

Split $R_E$ into an unbypassed $R_{E1}$ in series with a bypassed $R_{E2}$. The classical result, and its exact form:

$$A_v \approx -\frac{R_{ac}}{r_e + R_{E1}} \quad \text{(approximate)}, \qquad \boxed{A_v = -\frac{\beta\, R_{ac}}{r_\pi + (\beta+1) R_{E1}}} \quad (\text{exact, } r_o \to \infty) \tag{13}$$

With $R_{E1} = 100\,\Omega$ in Design A these give $-16.27$ and $-16.14$ respectively — the simple formula carries its own 0.8% error, which is *larger* than the Early-effect error we are about to compute.

Including $r_o$ is subtle here, because emitter degeneration **boosts** the output resistance looking into the collector:

$$r_{o,\text{eff}} = r_o\,[1 + g_m (R_{E1} \parallel r_\pi)] = 88.2\text{k} \times 5.18 = 457\,\text{k}\Omega. \tag{14}$$

The Early error therefore falls to $R_{ac}/r_{o,\text{eff}} = 2\text{k}/457\text{k} = \mathbf{0.46}\%$ (exact solution: $-16.14 \to -16.06$).

**Observation**

Degeneration makes the stage *less* sensitive to the Early effect, not more. The same feedback that stabilises the gain against $\beta$, temperature and $r_e$ also stabilises it against $r_o$.

### 3.6 Summary of the gain equations

| Case | Expression | Design A | Error |
|---|---|---|---|
| 1. Ideal, bypassed | $-g_m(R_C \parallel R_L)$ | −87.18 | +2.27% |
| 2. With Early | $-g_m(r_o \parallel R_C \parallel R_L)$ | −85.24 | — |
| 3. Referred to source | $A_v R_i/(R_i + R_s)$ | −63.81 | — |
| 4a. Swamped, approx | $-R_{ac}/(r_e + R_{E1})$ | −16.27 | +1.28% |
| 4b. Swamped, exact ideal | $-\beta R_{ac}/[r_\pi + (\beta{+}1)R_{E1}]$ | −16.14 | +0.46% |
| 4c. Swamped, with Early | $r_o \to r_o[1 + g_m(R_{E1} \parallel r_\pi)]$ | −16.06 | — |
| 5. Unloaded ($R_L$ removed) | $-g_m(r_o \parallel R_C)$ | −166.9 | +4.5% |
| 6. Ideal current-source load | $-g_m r_o = -V_A/V_T$ | −3846 | $\infty$ |

Line 6 is the important one, and it is the subject of Section 7.

## 4 Moving the Q-Point I: Fixed $I_C$, Varying $R_C$

Hold the bias network of Design A exactly as it is, so $I_C = 1.133\,\text{mA}$ is fixed, and sweep $R_C$. From Eq. (2), $V_{CE} = 10.3 - 1.133\text{m}\,R_C$, so increasing $R_C$ slides the Q-point down the load line toward saturation.

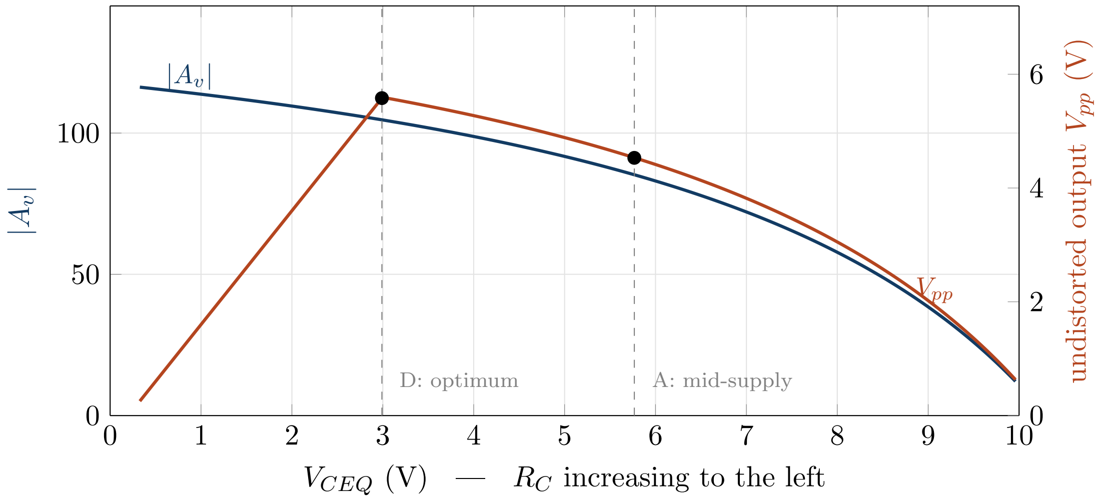


Figure 3: Sweeping $R_C$ at fixed bias current. Gain rises monotonically as the Q-point moves toward saturation (left), but swing peaks at $V_{CE} \approx 3.0\,\text{V}$ and then collapses. Mid-supply (Design A) sits well to the right of the swing optimum.

**Observations from Figure 3**

1. **Gain rises steadily to the left.** This is Eq. (7) at work: moving left means a larger $V_{RC}$, and gain is proportional to it.
2. **Swing has a clear maximum** at $V_{CE} = 3.0\,\text{V}$ ($R_C = 6.45\,\text{k}\Omega$), exactly where Eq. (5) predicted. To the right of it the amplifier is cutoff-limited; to the left it is saturation-limited.
3. **Between mid-supply and the optimum, both quantities improve simultaneously.** Moving from Design A to Design D raises the gain from 85 to 105 *and* the swing from 4.53 to 5.58 $V_{pp}$. Mid-supply is simply not on the Pareto front for an AC-coupled stage.
4. **Past the optimum there is a genuine trade.** Beyond $V_{CE} \approx 3\,\text{V}$ every extra decibel of gain is paid for in headroom, and beyond $V_{CE} \approx 1.5\,\text{V}$ the amplifier becomes useless

despite its impressive small-signal gain.

## 5 Moving the Q-Point II: Fixed Q-Point, Trading $R_C$ Against $I_C$

Now hold the Q-point itself fixed at Design A's position ($V_{CE} = 5.77\,\text{V}$, so $V_{RC} = 4.53\,\text{V}$) and slide along the hyperbola $I_C R_C = 4.53\,\text{V}$. Since $V_{RC}$ is constant, Eq. (7) says the *unloaded* gain is constant — but the loaded gain is not, because of the $R_L/(R_C + R_L)$ factor in Eq. (8).

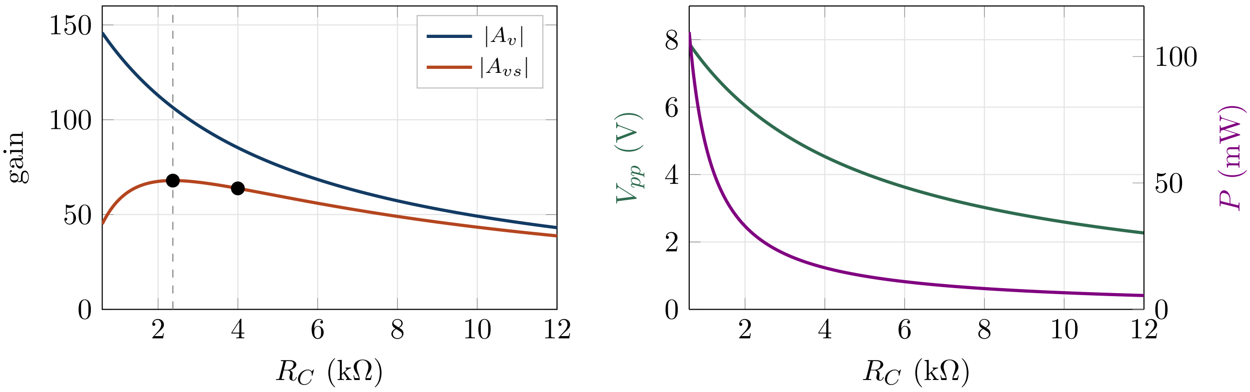


Figure 4: Same Q-point, different $R_C/I_C$ split. Left: raw gain climbs steeply as $R_C$ falls, but gain *from the source* peaks near $R_C = 2.4\,\text{k}\Omega$ and then collapses. Right: swing improves and power consumption explodes.

| $R_C$ | $I_C$ | $r_\pi$ | $R_i$ | $\lvert A_v\rvert$ | $\lvert A_{vs}\rvert$ | $P$ |
|---|---|---|---|---|---|---|
| 0.6 kΩ | 7.56 mA | 344 Ω | 268 Ω | 145.9 | 45.0 | 110 mW |
| 1.5 kΩ | 3.02 mA | 861 Ω | 670 Ω | 123.1 | 64.6 | 44 mW |
| **2.4 kΩ** | 1.91 mA | 1.36 kΩ | 1.06 kΩ | **106.4** | **67.9** | 28 mW |
| 4.0 kΩ | 1.13 mA | 2.29 kΩ | 1.79 kΩ | 85.2 | 63.8 | 16 mW |
| 10 kΩ | 0.45 mA | 5.74 kΩ | 4.47 kΩ | 49.2 | 43.4 | 6.6 mW |

**Observations from Figure 4**

1. **Raw gain and delivered gain disagree sharply.** Dropping $R_C$ from 4 kΩ to 0.6 kΩ raises $|A_v|$ by 71% but *lowers* $|A_{vs}|$ by 30%. The extra $g_m$ is entirely consumed by the collapse of $r_\pi$, and hence of $R_i$, which throws the signal away in $R_s$ before it ever reaches the base.

2. **There is a genuine optimum** near $R_C \approx 2.4\,\text{k}\Omega \approx 0.6 R_L$. It is broad and flat — anything from 2 to 3 kΩ is within 1% — so this is a robust design choice rather than a knife edge.

3. **The optimum costs power.** 28 mW versus 16 mW for a 6% gain improvement. In a battery-powered design you would stay at 4 kΩ; on a bench supply you would not.

4. **Swing improves monotonically as $R_C$ falls**, because $R_{ac}$ falls more slowly than $I_C$ rises. If undistorted output amplitude is the specification that matters, push $R_C$ low.

5. **The classic rule of thumb "make $R_C$ comparable to $R_L$" is vindicated**, but the reason is the interaction of three separate effects, not one.

## 6 Where the Early Effect Actually Matters

Everything so far has made $r_o$ look like an irrelevance — a 1 to 4% correction. That conclusion is correct *for a resistively loaded stage*, and completely wrong in general.

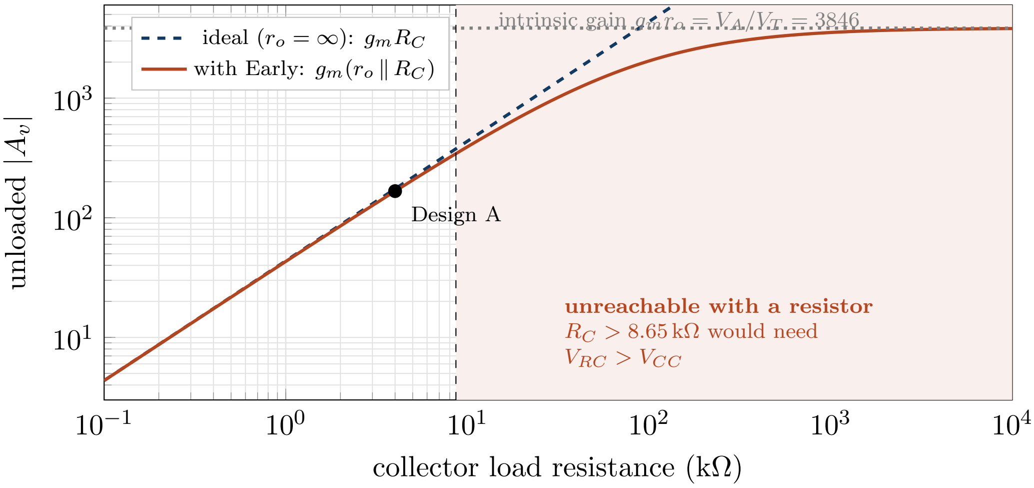


Figure 5: Unloaded gain versus collector load at fixed $I_C = 1.13\,\mathrm{mA}$. The two curves are indistinguishable throughout the region a resistor can actually reach.

Two facts collide in Figure 5.

**Fact one: the intrinsic gain is bias-independent.** Multiply $g_m$ by $r_o$:

$$g_m r_o = \frac{I_C}{V_T} \cdot \frac{V_A}{I_C} = \boxed{\frac{V_A}{V_T}} = \frac{100}{0.026} = 3846 \tag{15}$$

The $I_C$ cancels. This is the *maximum voltage gain the transistor can ever deliver*, a property of the device alone, and no choice of operating current changes it.

**Fact two: a resistor can never get you there.** To reach $|A_v| = 3846$ you would need $R_C \gg r_o = 88\,\mathrm{k\Omega}$, which at 1.13 mA would drop over 100 V across the resistor. The supply budget of Eq. (2) caps the usable $R_C$ at $8.65\,\mathrm{k\Omega}$, and the corresponding gain at 377 — ten times below the intrinsic ceiling. The shaded region of Figure 5 is forbidden by DC physics, not by the transistor.

**Why integrated circuits care about $r_o$ and your breadboard does not**

The escape is to replace $R_C$ with a **current source**, which presents a huge *small-signal* resistance while dropping only a volt or so of *DC*. This decouples the two facts above: the load resistance can be hundreds of kilohms while $V_{RC}$ stays small.

At that point $R_C \to \infty$ and Eq. (10) collapses to $A_v = -g_m r_o = -V_A/V_T$. The Early effect is no longer a correction — **it is the only thing setting the gain**. This is precisely why every analogue IC textbook makes such a fuss about $V_A$ while a discrete-design course can almost ignore it.

Rule: *$r_o$ matters when the collector load is comparable to it.* With resistors it never is; with active loads it always is.

Two further places $r_o$ shows up even in discrete work:

- **Output resistance.** $R_{out} = R_C \| r_o = 4 \| 88.2 = 3.83\,\mathrm{k\Omega}$ rather than $4\,\mathrm{k\Omega}$. A 4% correction, but it is the quantity that matters when cascading stages.
- **Distortion.** $r_o$ varies with the instantaneous $v_{CE}$ across the swing, so it contributes a little second-harmonic distortion — swamped, in practice, by the far larger nonlinearity of $g_m = i_C/V_T$.

**The Early effect and the DC bias point**

Strictly, $I_C = I_S e^{V_{BE}/V_T}(1+V_{CE}/V_A)$, so the Q-point current depends on $V_{CE}$. In *this* topology that dependence is almost invisible, because the divider and $R_E$ fix $I_E \approx (V_B - V_{BE})/R_E$ regardless of what the transistor would prefer. The negative feedback through $R_E$ absorbs it. In a fixed-base-current bias scheme, by contrast, the same Early effect tilts the whole load line and shifts the Q-point measurably — one more reason emitter degeneration is the standard choice.

# 7 Seven Designs Compared

All seven share the bias philosophy of Section 1. Designs A–F have a fully bypassed emitter; G is Design A with 100 Ω left unbypassed.

| | **A** mid-supply | **B** toward sat. | **C** toward cutoff | **D** opt. swing | **E** low $R_C$ | **F** high $R_C$ | **G** swamped |
|---|---|---|---|---|---|---|---|
| $I_C$ (mA) | 1.13 | 1.80 | 0.60 | 1.13 | 3.02 | 0.45 | 1.13 |
| $R_C$ (kΩ) | 4.0 | 4.0 | 4.0 | 6.45 | 1.5 | 10 | 4.0 |
| $R_E$ (Ω) | 1500 | 944 | 2833 | 1500 | 563 | 3753 | 100+1400 |
| $R_1 \| R_2$ (kΩ) | 40 / 10 | 25 / 6.4 | 76 / 19 | 40 / 10 | 15 / 3.8 | 101 / 25 | 40 / 10 |
| $V_{CE}$ (V) | 5.77 | 3.10 | 7.90 | 2.99 | 5.77 | 5.77 | 5.77 |
| $V_{RC}$ (V) | 4.53 | 7.20 | 2.40 | 7.31 | 4.53 | 4.53 | 4.53 |
| $g_m$ (mS) | 43.6 | 69.2 | 23.1 | 43.6 | 116 | 17.4 | 43.6 |
| $r_\pi$ (kΩ) | 2.29 | 1.44 | 4.33 | 2.29 | 0.86 | 5.74 | 2.29 |
| $r_o$ (kΩ) | 88.2 | 55.6 | 167 | 88.2 | 33.1 | 221 | 457* |
| $R_i$ (kΩ) | 1.79 | 1.13 | 3.37 | 1.79 | 0.67 | 4.47 | 4.89 |
| $\lvert A_v\rvert$ ideal | 87.2 | 138.5 | 46.2 | 107.6 | 126.7 | 49.8 | 16.14 |
| $\lvert A_v\rvert$ with Early | **85.2** | **133.7** | **45.6** | **104.7** | **122.7** | **49.1** | **16.06** |
| Early error | 2.27% | 3.60% | 1.20% | 2.80% | 3.30% | 1.29% | 0.46% |
| $\lvert A_{vs}\rvert$ | 63.8 | 87.2 | 38.7 | 78.4 | 64.7 | 43.3 | 14.3 |
| $R_{out}$ (kΩ) | 3.83 | 3.73 | 3.91 | 6.01 | 1.44 | 9.57 | 3.97 |
| $V_{pp}$ (V) | 4.53 | 5.80 | 2.40 | 5.58 | **6.59** | 2.59 | 4.53 |
| limited by | cutoff | sat. | cutoff | sat. | sat. | cutoff | cutoff |
| $f_L$ (Hz) | 285 | 411 | 165 | 285 | 601 | 128 | **68** |
| $f_H$ (MHz) | 0.95 | 0.70 | 1.52 | 0.78 | 0.90 | 1.39 | **3.04** |
| GBW (MHz) | 80.6 | 93.5 | 69.2 | 82.0 | 110 | 68.1 | 48.8 |
| $P$ (mW) | 16.5 | 26.1 | 8.7 | 16.5 | 43.9 | 6.6 | 16.5 |

* For G this is the *effective* output resistance $r_o[1 + g_m(R_{E1} \| r_\pi)]$, boosted 5.2× by degeneration.

## 7.1 Reading the table

**Design-by-design**

**A — mid-supply.** The safe textbook answer, and not the best at anything. Its only real virtue is that $V_{CE}$ is far from both rails, so the design tolerates component drift and temperature swing without approaching either clipping mechanism.

**B — pushed toward saturation.** The highest gain of the fully bypassed set (134) *and* a good swing (5.8 $V_{pp}$), because higher $I_C$ pushes the cutoff limit up while the lower $V_{CE}$ happens to land near the AC optimum. The costs are 60% more power and only 2.9 V of margin to saturation — a 10% drift in $\beta$ or temperature could clip it.

**C — pushed toward cutoff.** Nearly the worst of everything: half the gain, half the swing, and the only saving is power. Sitting high on the load line is almost always a mistake in a voltage amplifier.

**D — AC-optimum swing.** What the classic rule *should* have given. More gain and more swing than A at identical power. If you take one design lesson from this document, it is that Eq. (5) beats $V_{CC}/2$ whenever the load is AC-coupled.

**E — low $R_C$, high $I_C$.** Best swing in the table (6.6 $V_{pp}$) and high raw gain, but $|A_{vs}|$ is barely better than A's because $R_i$ has collapsed to $670\,\Omega$. Nearly triple the power for a 1.4% improvement in delivered gain. A cautionary tale about optimising the wrong variable.

**F — high $R_C$, low $I_C$.** Frugal (6.6 mW) with a high input resistance, but low gain, poor swing, and a $9.6\,\mathrm{k}\Omega$ output resistance that will be crushed by any following stage. Suitable only when the next stage is a high-impedance buffer.

**G — swamped.** Gain drops by a factor of 5.3, and in exchange: input resistance nearly triples, upper cutoff triples to 3 MHz, lower cutoff drops to 68 Hz, the Early error nearly vanishes, and the gain becomes almost independent of $\beta$, $I_C$ and temperature.

## 8 What the Q-Point Does to the Frequency Response

Every design decision above quietly moved the cutoff frequencies computed in the companion document. The mechanisms are worth spelling out.

| Change | Mechanism | Effect |
|---|---|---|
| $I_C \uparrow$ | $r_e \downarrow$, so $R_{eq}$ seen by $C_E$ falls | $f_{LE} \uparrow$ (worse) |
| $I_C \uparrow$ | $r_\pi \downarrow$, so $R_i \downarrow$ | $f_{L1} \uparrow$ (worse) |
| $I_C \uparrow$ | $C_\pi = C_{je} + g_m \tau_F$ grows with $g_m$ | $f_{H1} \downarrow$ (worse) |
| $\lvert A_v \rvert \uparrow$ | Miller: $C_\mu(1 + \lvert A_v \rvert)$ grows | $f_{H1} \downarrow$ (worse) |
| $R_C \uparrow$ | $R_{ac} \uparrow$, larger output time constant | $f_{H2} \downarrow$ |
| $R_{E1} > 0$ | gain drops $\Rightarrow$ Miller drops; $R_i \uparrow$ | $f_H \uparrow\uparrow$, $f_L \downarrow$ |

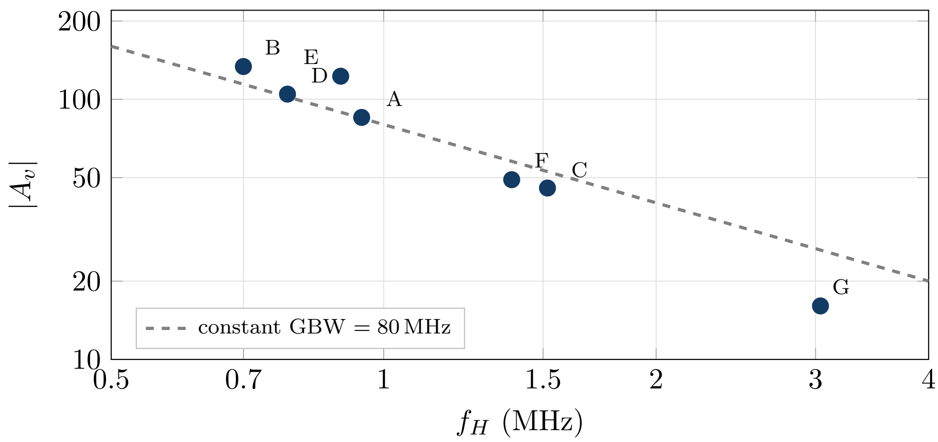


Figure 6: Gain versus upper cutoff for all seven designs. A–F cluster loosely around a constant gain–bandwidth hyperbola; the swamped design G falls well below it.

**Observation — gain–bandwidth is a guideline, not a law**

Designs A–F span a 3:1 range of gain, yet their GBW products stay between 68 and 110 MHz. That is the familiar constant-GBW behaviour, and it holds because Miller multiplication makes $C_{in} \propto |A_v|$ while $R_{th,in}$ barely changes.

Design G breaks it, at 49 MHz. Degeneration cuts the gain by 5.3× but raises $f_H$ by only 3.2×, because the $C_\pi$ term in $C_{in}$ does not shrink with the gain — once Miller has been beaten down to 70 pF, the 30 pF of $C_\pi$ starts to matter. **Trading gain for bandwidth via degeneration is real but sub-linear**; the cascode, which removes Miller without sacrificing $g_m$, does better.

## 9 Design Rules Extracted

**Sequence for designing this stage**

1. **Fix $V_E$ first** at 1–2 V (here 1.7 V). This buys bias stability and costs only headroom you can afford.
2. **Choose $R_C$ near** 0.5–1 × $R_L$, not much larger. Larger $R_C$ throws signal away in the $R_C/R_L$ divider and raises $R_{out}$.
3. **Place $V_{CE}$ using the AC load line**, $V_{CE} \approx V_{CC}R_{ac}/(R_{ac} + R_{dc})$, not at $V_{CC}/2$ — unless the load is DC-coupled, in which case $R_{ac} = R_{dc}$ and the two rules coincide.
4. **Set $I_C$ from the swing requirement**, $I_C \geq v_{o,peak}/R_{ac}$, then check the power.
5. **Check $R_i$ against $R_s$.** If $R_i$ has fallen near $R_s$, extra $g_m$ is being wasted; back off $I_C$.
6. **Add $R_{E1}$ last**, to trade surplus gain for linearity, bandwidth, input resistance and predictability. Choose $R_{E1}$ so that $g_m R_{E1}$ is between 3 and 10.
7. **Ignore $r_o$** for a resistively loaded stage — but check $R_{ac}/r_o$ once to confirm it is a few percent. If you ever move to an active load, $r_o$ becomes the whole story.

**Five traps**

1. **Designing on the DC load line.** The signal never travels on it. Always draw the AC load line before quoting a swing.
2. **Chasing $g_m$.** Raising $I_C$ raises $g_m$ but lowers $r_\pi$, $R_i$ and $f_H$, and raises $f_L$. Design E gains 44% raw and 1.4% delivered.
3. **Quoting $A_v$ when the specification is $A_{vs}$.** Always carry the $R_i/(R_i + R_s)$ factor through.
4. **Assuming the emitter is fully bypassed at all frequencies.** It is bypassed only above $f_{LE}$; below it, the stage silently becomes Design G.
5. **Using $-R_{ac}/(r_e + R_{E1})$ and then worrying about the Early effect.** For Design G that approximation is nearly three times more wrong than $r_o$ is.

**Self-test**

1. A stage has $V_{RC} = 6$ V and no external load. Estimate $|A_v|$ without computing $g_m$ or $R_C$ separately.
2. The same stage is built with a transistor of $V_A = 40$ V instead of 100 V, at $I_C = 2$ mA with $R_{ac} = 3\,\text{k}\Omega$. What percentage error do you make by ignoring $r_o$?
3. Why does raising $I_C$ at fixed $R_C$ move the Q-point toward saturation rather than cutoff?
4. Design D has more gain *and* more swing than Design A at the same power. Where did the improvement come from — what was Design A wasting?
5. A colleague proposes reaching $|A_v| = 2000$ from one CE stage on a 12 V supply by using

a 500 kΩ collector resistor. What goes wrong, and what would you use instead?

**Answers**

1. $|A_v| = V_{RC}/V_T = 6/0.026 \approx 231$.
2. $r_o = 40/2\text{m} = 20\,\text{k}\Omega$; error $= R_{ac}/r_o = 3/20 = \mathbf{15\%}$ — no longer negligible. Low $V_A$ combined with high $I_C$ and a large $R_{ac}$ is exactly the corner where the ideal model fails.
3. Because $V_{CE} = 10.3 - I_C R_C$: more current means a larger drop across $R_C$, leaving less across the transistor. Saturation is the low-$V_{CE}$ end.
4. Headroom on the saturation side. Design A could fall 5.57 V but only rise 2.27 V, so 3.3 V of DC bias was doing nothing. D moves the Q-point down until both limits are equal, and the $R_C$ increase that achieves this also raises $V_{RC}$ and hence the gain.
5. At $I_C = 1.13\,\text{mA}$ a 500 kΩ resistor would need 565 V across it. The DC budget of Eq. (2) caps $R_C$ at about 8.6 kΩ. Even if it were possible, $r_o = 88\,\text{k}\Omega$ would clamp the gain to $g_m r_o = 3846$ at best. The correct answer is an active (current-source) load, which presents a high AC resistance with a small DC drop; or two cascaded lower-gain stages.

# A Appendix — Formula Sheet

| | |
|---|---|
| DC budget | $V_{RC} + V_{CE} = V_{CC} - V_E$ |
| Bias | $R_E = V_E/I_C, \quad R_1{+}R_2 = 12\beta/(21 I_C), \quad R_1{:}R_2 = 4{:}1$ |
| Load lines | $R_{dc} = R_C + R_E, \quad R_{ac} = R_C \parallel R_L$ |
| Clipping limits | $v_o^+ = I_C R_{ac}$ (cutoff), $v_o^- = V_{CE} - V_{CE,sat}$ (saturation) |
| Optimum Q | $V_{CE,opt} \approx V_{CC} R_{ac}/(R_{ac} + R_{dc})$ |
| Small-signal | $g_m = I_C/V_T, \quad r_e = V_T/I_C, \quad r_\pi = \beta r_e, \quad r_o = V_A/I_C$ |
| Gain, ideal | $A_v = -g_m R_{ac}$ |
| Gain, with Early | $A_v = -g_m (r_o \parallel R_C \parallel R_L)$ |
| Unloaded gain | $\lvert A_v \rvert = V_{RC}/V_T$ |
| Loaded gain | $\lvert A_v \rvert = (V_{RC}/V_T) \cdot R_L/(R_C + R_L)$ |
| Source-referred | $A_{vs} = A_v R_i/(R_i + R_s), \quad R_i = R_1 \parallel R_2 \parallel r_\pi$ |
| Swamped, exact | $A_v = -\beta R_{ac}/[r_\pi + (\beta{+}1) R_{E1}]$ |
| Swamped $r_o$ boost | $r_{o,\text{eff}} = r_o[1 + g_m(R_{E1} \parallel r_\pi)]$ |
| Output resistance | $R_{out} = R_C \parallel r_{o,\text{eff}}$ |
| Early error | $R_{ac}/r_o = I_C R_{ac}/V_A = v_o^+/V_A$ |
| Intrinsic gain | $g_m r_o = V_A/V_T$ (bias-independent) |
| Resistor-load ceiling | $\lvert A_v \rvert_{\max} \approx (V_{CC} - V_E - V_{CE,\min})/V_T$ |

*End of document.*